

Vectors: Basic concepts (Lec 1-3)

Lec 1:- We recalled the notions of vectors & scalars with examples. We saw that two vectors in $\mathbb{R}^n$ are equal regardless of their initial & terminal points if they have the same magnitude & direction.

We saw the (triangular law & parallelogram law) addition operation in $\mathbb{R}^n$ & defined zero vector. Further, we defined the unique negative (inverse) of a vector. We saw that $(\mathbb{R}^n, +)$ forms an abelian group under vector addition.

Further, we saw scalar multiplication (by real no.) on $\mathbb{R}^n$ & by noting the distributive properties, we saw that $\mathbb{R}^n$ becomes a vector space over the field of real numbers $\mathbb{R}$. (We demonstrated these vector space properties geometrically in the case $n=3$.)

Further, we recalled the Cartesian Coordinate system XYZ in $\mathbb{R}^3$ & saw that if $P = (x_1, y_1, z_1)$ & $Q = (x_2, y_2, z_2)$ are two points in $\mathbb{R}^3$ w.r.t this co-ordinate system, then $x_2 - x_1, y_2 - y_1$ & $z_2 - z_1$ are called the components of the vector $\vec{PQ}$ in the X, Y & Z directions respectively.

Further, we defined the length or norm of the vector $\vec{PQ}$ as

$$\|\vec{PQ}\| := \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Notice that we use the symbol $\|\cdot\|$ instead of the usual symbol $|\cdot|$ to distinguish it from the absolute value symbol $|\cdot|$. This will keep us in distinguishing between lengths of vectors & lengths of scalars (which is their absolute value!).

for eg. if $\vec{v} = -2(\hat{i} + 2\hat{j})$, then

$$\|\vec{v}\| = |-2| \cdot \|\hat{i} + 2\hat{j}\|$$

$$\text{i.e., } \|\vec{v}\| = 2 \times \sqrt{5}$$

Inner product :-

What differentiates $\mathbb{R}^n$ from an arbitrary n dimensional vector space is the existence of concepts like inner products & angle between two vectors.

For e.g., it is not a priori understood what is the length of any polynomial in following 3d vector space

$$V = \{a_0 + a_1x + a_2x^2 \mid a_0, a_1, a_2 \in \mathbb{R}\}$$

Let us take the case $\mathbb{R}^3$. Here, inner product or dot product of two vectors $\vec{a}$ & $\vec{b}$ is

$$\vec{a} \cdot \vec{b} = \begin{cases} \|\vec{a}\| \cdot \|\vec{b}\| \cos \theta & \text{if } \vec{a} \neq 0 \text{ \& } \vec{b} \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\theta \in [0, \pi]$ is the acute angle between $\vec{a}$ & $\vec{b}$.

If we use the Cartesian coordinates

$$\vec{a} = (a_1, a_2, a_3)$$

$$\vec{b} = (b_1, b_2, b_3), \text{ then}$$

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

Orthogonality

A vector $\vec{a}$ is orthogonal to $\vec{b}$ if $\vec{a} \cdot \vec{b} = 0$. Since inner product commutes, hence, we can say $\vec{b}$ is also orthogonal to $\vec{a}$.

i.e., $\vec{a}$ & $\vec{b}$ are orthogonal to each other.

Theorem

The inner product of two non-zero vectors is 0 iff these vectors are $\perp$.

The necessary & sufficient condition for two non-zero vectors to be orthogonal is that they are perpendicular.

Proof:- " $\Rightarrow$ " (only if part) (or necessary part)

Suppose $\vec{a} \cdot \vec{b} = 0$, i.e., $\|\vec{a}\| \cdot \|\vec{b}\| \cos \theta = 0$

$\Rightarrow \theta = \pi/2$ as $\vec{a}$ & $\vec{b}$ are non zero vectors

$\Leftarrow$ Let $\vec{a}$ & $\vec{b}$ be $\perp$ & non-zero.

(if part) Then, $\vec{a} \cdot \vec{b} = \|\vec{a}\| \cdot \|\vec{b}\| \cos \pi/2 = 0$

(sufficient part)

Length & angle

$$\|\vec{a}\| = \sqrt{\vec{a} \cdot \vec{a}}$$

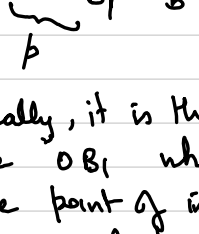
$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \cdot \|\vec{b}\|} \quad \left(\text{only defined if both } \vec{a} \text{ \& } \vec{b} \text{ are non zero} \right)$$

Cauchy-Schwarz inequality:- $|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\| \|\vec{b}\|$

Triangle inequality

$$\|\vec{a} + \vec{b}\| \leq \|\vec{a}\| + \|\vec{b}\|$$

Projection of $\vec{a}$ in the direction of $\vec{b}$



Projection of $\vec{a}$ along a non zero vector $\vec{b}$ is

$$p = \|\vec{a}\| \cos \theta$$

$$= \vec{a} \cdot \left(\frac{\vec{b}}{\|\vec{b}\|} \right)$$

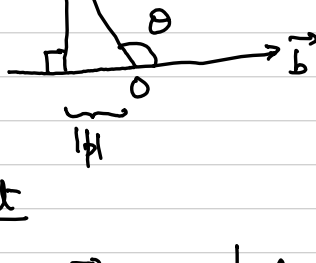
$$= \frac{\vec{a} \cdot \vec{b}}{\|\vec{b}\|}$$

Geometrically, it is the distance OB_1 where B_1 is the point of intersection of the normal to the vector $\vec{b}$ passing through terminal point of vector $\vec{a}$.

Some also consider projection as a vector. Then,

$$\vec{OB_1} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{b}\|^2} \vec{b} = \left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{b}\|^2} \right) \vec{b}$$

If θ is between 90° & 180° , then projection p is a -ve value. as it is antiparallel to $\vec{b}$.



Vector Product

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

(is a vector)

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \cdot \|\vec{b}\| \sin \theta$$

Direction: determined by right hand screw rule.



Scalar & vector fields

Let $f: X \rightarrow Y$

X : domain of f

Y : co domain of f

We say f is a scalar function or scalar valued function if $f(P)$ is a scalar for every P in X . We also say that f defines a scalar field in that domain of definition of f .

Eg. a) Temperature field of a body or at any point on the Earth's surface.

b) Pressure field of the air in Earth's atmosphere.

c) Any function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ or $f: \mathbb{R}^3 \rightarrow \mathbb{R}$.

$$\text{like } f(x, y, z) = x^2y - z^2$$

d) distance of any place in earth from your home.

$$P = (x, y, z)$$

$$H = (a, b, c)$$

$$f(P) = \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2}$$

e) Divergence of a vector field defines a scalar field. (to be defined later).

Vector fields :-

We say v is a vector function i.e. whose values $v(P)$ (for P in the domain of v) are vectors in the 3 dimensional space.

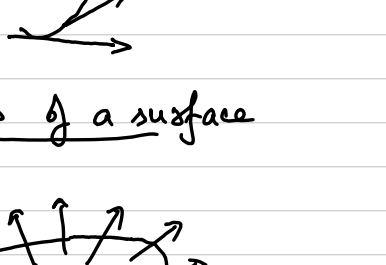
$$v = v(P) = (v_1(P), v_2(P), v_3(P))$$

A vector function v defines a vector field in its domain of definition.

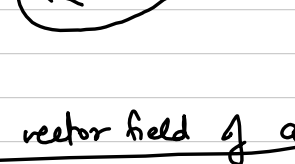
If $v: \mathbb{R}^n \rightarrow \mathbb{R}^m$, then we call v as a vector valued function of a vector variable. (usually when $n, m > 1$).

Typical domains in applications are either $\mathbb{R}^3$ or a surface $z = f(x, y)$ in $\mathbb{R}^3$ or a curve in $\mathbb{R}^3$.

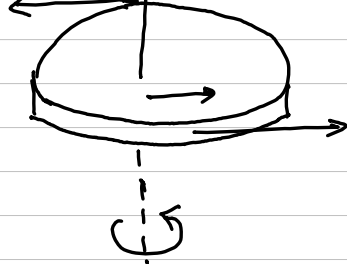
Eg. a) Field of tangent vectors of a smooth curve



b) Normal vectors of a surface



c) Velocity vector field of a rotating body

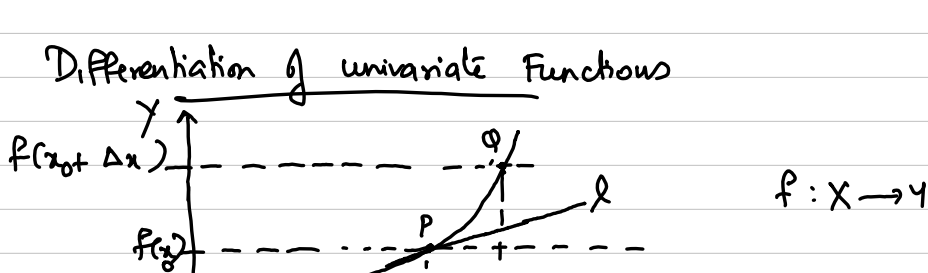


d) Gravitational force field of earth on any object

e) Gradient of a scalar field defines a vector field

f) Curl of a differentiable vector field defines another vector field.

Differentiation of univariate Functions



The slope of the secant line PQ is $\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$

As $\Delta x \rightarrow 0$ (i.e. as Q approaches P), the secant line will approach the tangent line t &

It $\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ will be the slope of t .

Thus, $f'(x_0)$ is the slope of the tangent to f at P .

Here, we assume the above limit exists.

We will see the notion of derivatives for vector valued functions in upcoming lectures.

Vector Calculus

Let $\vec{a}_n$ be an infinite sequence of vectors in $\mathbb{R}^3$.

Further, $\vec{a}_n = a_{n,1}\hat{i} + a_{n,2}\hat{j} + a_{n,3}\hat{k}$.

We say $\vec{a}_n$, $n \in \{1, 2, \dots\}$ converges if $\exists \vec{a} \in \mathbb{R}^3$ such that

$$\lim_{n \rightarrow \infty} \|\vec{a}_n - \vec{a}\| = 0.$$

Here, $\|\vec{a}\| = \|x_1\hat{i} + x_2\hat{j} + x_3\hat{k}\| \stackrel{\text{def}}{=} \sqrt{x_1^2 + x_2^2 + x_3^2}$ (called ℓ_2 norm)

$\vec{a}$ is called the limit vector of $\vec{a}_n$ & we write

$$\lim_{n \rightarrow \infty} \vec{a}_n = \vec{a}.$$

$$\begin{aligned} \text{Thm 1) } \vec{a}_n \rightarrow \vec{a} &\Leftrightarrow \lim_{n \rightarrow \infty} a_{n,1} = a_1, \\ &\lim_{n \rightarrow \infty} a_{n,2} = a_2 \\ &\& \lim_{n \rightarrow \infty} a_{n,3} = a_3. \end{aligned}$$

Proof:- " $\Rightarrow$ "

$$\text{Given: } \lim_{n \rightarrow \infty} \|\vec{a}_n - \vec{a}\| = 0$$

This implies $\lim_{n \rightarrow \infty} \|\vec{a}_n - \vec{a}\|^2 = 0$ as $\|\vec{a}_n - \vec{a}\|^2 \leq \|\vec{a}_n - \vec{a}\|$ for n sufficiently large

$$\|\vec{a}_n - \vec{a}\|^2 = |a_{n,1} - a_1|^2 + |a_{n,2} - a_2|^2 + |a_{n,3} - a_3|^2 = 0$$

In particular, for each $r \in \{1, 2, 3\}$

$$|a_{n,r} - a_r|^2 \leq \|\vec{a}_n - \vec{a}\|^2$$

$$\therefore \lim_{n \rightarrow \infty} |a_{n,r} - a_r| \leq \lim_{n \rightarrow \infty} \|\vec{a}_n - \vec{a}\|$$

Thus,

$$\lim_{n \rightarrow \infty} a_{n,r} = a_r \text{ for each } r \in \{1, 2, 3\}.$$

" $\Leftarrow$ " Applying limits on both sides of $\sqrt{(\quad)}$, we see that

$$\begin{aligned} \lim_{n \rightarrow \infty} \|\vec{a}_n - \vec{a}\| &= \lim_{n \rightarrow \infty} \sqrt{(|a_{n,1} - a_1|^2 + |a_{n,2} - a_2|^2 + |a_{n,3} - a_3|^2)} \\ &= \left(\lim_{n \rightarrow \infty} |a_{n,1} - a_1|^2 + \lim_{n \rightarrow \infty} |a_{n,2} - a_2|^2 + \lim_{n \rightarrow \infty} |a_{n,3} - a_3|^2 \right)^{1/2} \\ &\leq \left(\lim_{n \rightarrow \infty} |a_{n,1} - a_1| + \lim_{n \rightarrow \infty} |a_{n,2} - a_2| + \lim_{n \rightarrow \infty} |a_{n,3} - a_3| \right)^{1/2} \\ &\because |a_{n,r} - a_r|^2 \leq |a_{n,r} - a_r| \text{ for } n \text{ suff. large} \\ &= 0. \end{aligned}$$

Another way:- $\Leftarrow$

$$\| (a_{n,1} - a_1)\hat{i} + (a_{n,2} - a_2)\hat{j} + (a_{n,3} - a_3)\hat{k} \|$$

$$\leq |a_{n,1} - a_1| + |a_{n,2} - a_2| + |a_{n,3} - a_3| \quad (\Delta \text{ inequality})$$

Applying limit $\lim_{n \rightarrow \infty}$ on both sides, we get

$$\lim_{n \rightarrow \infty} \|\vec{a}_n - \vec{a}\| \leq 0 \quad \& \text{ we are done.}$$

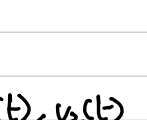
Open-Balls in $\mathbb{R}^n$, $n \geq 1$.

Let $\vec{a} \in \mathbb{R}^n$ & let $r > 0$. An open n -ball of radius r & center $\vec{a}$ is given by

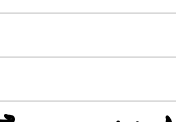
$$B(\vec{a}, r) = \{ \vec{x} \in \mathbb{R}^n \mid \|\vec{x} - \vec{a}\| < r \}.$$

$n=1$: Open intervals $(a-r, a+r)$

$n=2$: Open (circular) disks



$n=3$: Open spherical solids



Limit of a function

($S \subseteq \mathbb{R}$)

A vector function $v(t)$ of a real variable (say, $v: S \rightarrow \mathbb{R}^3$) is said to have the limit ℓ as $t \rightarrow t_0$

ie, $\lim_{t \rightarrow t_0} v(t) = \ell$ if

1) t_0 is a limit point of S

2) $v(t)$ is defined in $B(t_0, r) \setminus \{t_0\}$ for some $r > 0$ &

3) $\lim_{t \rightarrow t_0} \|v(t) - \ell\| = 0$.

Note: ℓ is a vector here

Continuity

A vector function $v(t)$ of a real variable (say, $v: S \rightarrow \mathbb{R}^3$) is continuous at $t_0 \in S$ (t_0 is a limit point of S) if

1) v is defined in $B(t_0, r)$ for some $r > 0$ &

2) $\lim_{t \rightarrow t_0} v(t) = v(t_0)$.

(Verify) v is continuous at $t_0 \Leftrightarrow v_1(t), v_2(t), v_3(t)$ are continuous at t_0 .

$$\text{where } v(t) = [v_1(t), v_2(t), v_3(t)].$$

Defn: Limit point

Let $S \subseteq \mathbb{R}^n$, $n \geq 1$. Assume $\vec{a} \in \mathbb{R}^n$. $\vec{a}$ is called a limit point of S if every neighbourhood of $\vec{a}$ contains a point of $S \setminus \{\vec{a}\}$. Here,

neighbourhood means an open n -ball $B(\vec{a}, r)$ of radius r around $\vec{a}$, and

In other words, limit $\lim_{t \rightarrow t_0} v(t)$ makes sense only if

t_0 is a limit point of S .

Derivative of Vector function (vector valued function)

Let $S \subseteq \mathbb{R}$.

A vector function $v: S \rightarrow \mathbb{R}^3$ is said to be differentiable at a point $t_0 \in S$ if

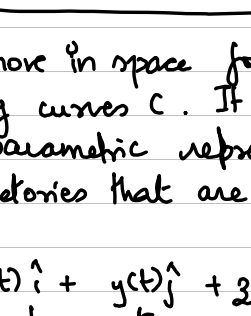
$$\lim_{h \rightarrow 0} \frac{v(t_0+h) - v(t_0)}{h} \text{ exists. Call } v'(t_0).$$

Equivalently,

$$\lim_{h \rightarrow 0} \left\| \frac{v(t_0+h) - v(t_0)}{h} - v'(t_0) \right\| = 0.$$

Here, $v'(t_0)$ is called the derivative of $v(t)$ at $t = t_0$.

Note that $v'(t)$ is also a vector valued function of a real variable t .



$v'(t_0)$ is called the tangent vector of the curve traced by the vector field v at $t = t_0$.

Ex 1) Show the following:- For u, v differentiable on $\mathbb{R}$ & c any scalar,

$$1) \frac{d}{dt}(u+v) = \frac{du}{dt} + \frac{dv}{dt}$$

$$2) \frac{d}{dt}(cv) = c \frac{dv}{dt} \quad c - \text{scalar}$$

Thus, we observe that on the vector space

$$V = \{ f: \mathbb{R} \rightarrow \mathbb{R}^3 \mid f \text{ is differentiable everywhere on } \mathbb{R} \}$$

the derivative map $\frac{d}{dt}$ is a linear map

Normal vector to a plane

Pr 1) Find a unit vector $\vec{n}$ to the plane $4x + 2y + 4z = -7$.

Further, find the distance of the plane from the origin.

Ans: Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ denote the position vector of any pt on the plane. Then, $\vec{r} \cdot \vec{a} = -7$, where $\vec{a} = 4\hat{i} + 2\hat{j} + 4\hat{k}$

Further, if $\vec{r}_1, \vec{r}_2$ denote the p.v. of any 2 pts say P & Q on the plane, then

$$(\vec{r}_2 - \vec{r}_1) \cdot \vec{a} = \vec{PQ} \cdot \vec{a} = 0 \quad (\text{Used linearity of dot product in the 2nd variable})$$

Thus, $\vec{a}$ is $\perp$ to any vector on the plane.

Thus, $\vec{a}$ is normal to this plane

$$\& \frac{\vec{a}}{\|\vec{a}\|} = \frac{4\hat{i} + 2\hat{j} + 4\hat{k}}{\sqrt{16+4+16}} = \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}.$$

b)

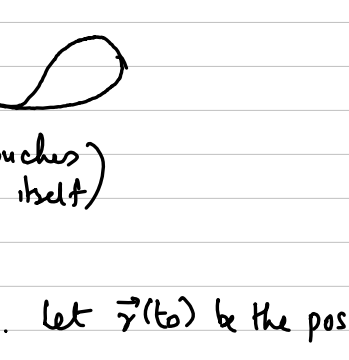
Let the normal from the origin to the plane hit the plane at Q , without loss of generality. Thus, $OQ \parallel \vec{a}$.

We need to find $\|\vec{OQ}\|$, ie, $|\vec{OP}| \cos \theta$

$$= \left| \vec{OP} \cdot \frac{\vec{a}}{\|\vec{a}\|} \right|$$

$$= \left| \vec{OP} \cdot \frac{\vec{a}}{\|\vec{a}\|} \right|$$

$$= \left| \frac{-7}{6} \right| = \frac{7}{6}.$$



Solve qn 1) using cross-products.

Curves in $\mathbb{R}^3$ & their parametric representation

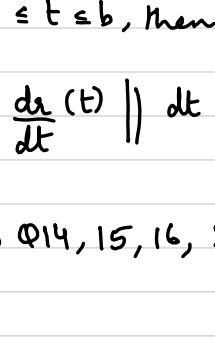
Bodies that move in space form paths that may be represented by curves C . It is often useful to use the following parametric representation for C . We will only study those trajectories that are "smooth" curves.

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

Here, t is the parameter

It is customary to write

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}.$$



Eg 1) The circle $x^2 + y^2 = 4$, $z = 0$ can be represented as

$$\vec{r}(t) = 2 \cos t \hat{i} + 2 \sin t \hat{j}$$

where $0 \leq t \leq 2\pi$.

$\vec{r}$ has CCW as orientation (in the positive sense).

The direction along the curve C which we traverse as t increases is called the orientation in the positive sense on C .

The other dirn. is called that in the negative sense.

Eg 2) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $z = 0$ is represented

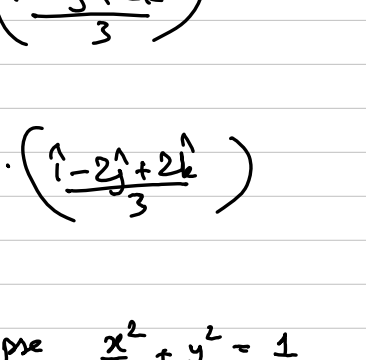
$$\text{as } \vec{r}(t) = a \cos t \hat{i} + b \sin t \hat{j} + 0 \hat{k}.$$

Eg 3) Straight line

A straight line through a point P with position vector $\vec{a}$ in the direction of a constant vector $\vec{b}$ can be represented parametrically by

$$\vec{r}(t) = \vec{a} + t\vec{b}$$

$$-\infty < t < \infty$$



Plane curve

A curve that lies in a plane in space.

Twisted curves

A curve that is not a plane curve is called a twisted curve.

Eg. 1. Circular helix

$$\vec{r}(t) = a \cos t \hat{i} + a \sin t \hat{j} + ct \hat{k}$$

It lies on the cylinder $x^2 + y^2 = a^2$.

$c > 0$
Right handed circular helix.

A simple curve is a curve without points at which the curve intersects or touches itself.

Non-examples

(intersects itself)

(touches itself)

Tangent to a smooth curve

Let the smooth curve C be given by $\vec{r}(t)$. Let $\vec{r}(t_0)$ be the posn. vector of P .

$$\text{Recall } \frac{d\vec{r}}{dt} \Big|_P = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t_0 + \Delta t) - \vec{r}(t_0)}{\Delta t}$$

If $\vec{r}'(t_0) \neq 0$, we call $\vec{r}'(t_0)$ as the tangent vector of the curve C at $P = \vec{r}(t_0)$.

Any point on the tangent vector at P , say Q , has the position vector $\vec{q}(w)$

$$\vec{q}(w) = \vec{r}(t_0) + w \vec{r}'(t_0)$$

Here, w is the parameter.

A curve $\vec{r}(t)$ is called smooth curve if $\vec{r}'(t)$ has a continuous derivative $\vec{r}''(t)$. If $a \leq t \leq b$, then

$$\text{Length of the smooth curve } C \stackrel{\text{defn}}{=} \int_a^b \left\| \frac{d\vec{r}}{dt}(t) \right\| dt$$

Work out Problem set 9.5 : Q11, 12, Q14, 15, 16, 29

(Kreyszig)

10th edition

International Student Version

Some problems

2) A particle moves along the curve (t -time)

$$x = 2t^2, y = t^2 - 4t, z = -t - 5$$

Find the components of its velocity & acceleration at $t = 1$ in the direction $\hat{i} - 2\hat{j} + 2\hat{k}$.

$$\text{Ans: } \vec{r}(t) = 2t^2\hat{i} + (t^2 - 4t)\hat{j} + (-t - 5)\hat{k}$$

$$\vec{v} = \frac{d\vec{r}}{dt}(1) = 4t\hat{i} + (2t - 4)\hat{j} - \hat{k} \Big|_{t=1}$$

$$= 4\hat{i} - 2\hat{j} - \hat{k}$$

$$\vec{a} = \frac{d^2\vec{r}}{dt^2}(1) = 4\hat{i} + 2\hat{j} \Big|_{t=1} = 4\hat{i} + 2\hat{j}$$

Components of $\vec{v}$ & $\vec{a}$ along $\hat{i} - 2\hat{j} + 2\hat{k} = \vec{b}$

$$\hat{b} = \frac{\hat{i} - 2\hat{j} + 2\hat{k}}{3} \quad (\text{Unit vector along } \vec{b})$$

$$\vec{v} \cdot \hat{b} \Big|_{t=1} = (4\hat{i} - 2\hat{j} - \hat{k}) \cdot \left(\frac{\hat{i} - 2\hat{j} + 2\hat{k}}{3} \right)$$

$$= 2.$$

$$\vec{a} \cdot \hat{b} \Big|_{t=1} = (4\hat{i} + 2\hat{j}) \cdot \left(\frac{\hat{i} - 2\hat{j} + 2\hat{k}}{3} \right)$$

$$= 0.$$

3) Find the tangent to the ellipse $\frac{x^2}{4} + y^2 = 1$

$$\text{at } P = (\sqrt{2}, \frac{1}{\sqrt{2}}).$$

$$\text{Ans: } \vec{r}(t) = 2 \cos t \hat{i} + \sin t \hat{j}$$

$$\vec{r}'(t) = -2 \sin t \hat{i} + \cos t \hat{j}$$

$$\text{Let } \vec{r}(t_0) = P \Rightarrow t_0 = \frac{\pi}{4}$$

$$\Rightarrow \vec{r}'(t_0) = -2 \sin \frac{\pi}{4} \hat{i} + \cos \frac{\pi}{4} \hat{j}$$

$$= -\sqrt{2}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$$

$$\Rightarrow \text{Tangent vector: } \vec{r} + w \vec{r}' = \sqrt{2}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + w(-\sqrt{2}\hat{i} + \frac{1}{\sqrt{2}}\hat{j})$$

Partial Derivatives of scalar fields :-

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$\frac{\partial f}{\partial x}(x_1, y_1)$: The partial deriv of f wrt x at (x_1, y_1) is

$$f_x(x_1, y_1) = \lim_{h \rightarrow 0} \frac{f(x_1 + h, y_1) - f(x_1, y_1)}{h} = \frac{\partial f}{\partial x}(x_1, y_1)$$

||⁴, the partial deriv of f wrt y at (x_1, y_1) is

$$f_y(x_1, y_1) = \lim_{k \rightarrow 0} \frac{f(x_1, y_1 + k) - f(x_1, y_1)}{k} = \frac{\partial f}{\partial y}(x_1, y_1).$$

The partial derivatives f_x & f_y are also functions of x & y . Geometric interpretation

Let $z = f(x, y)$ represent a surface in space.

The plane $y = y_1$ (vertical plane) cuts the surface, intersecting it along a curve & the partial derivative $\frac{\partial z}{\partial x}(x_1, y_1)$ represents the slope of the tangent to the curve.

